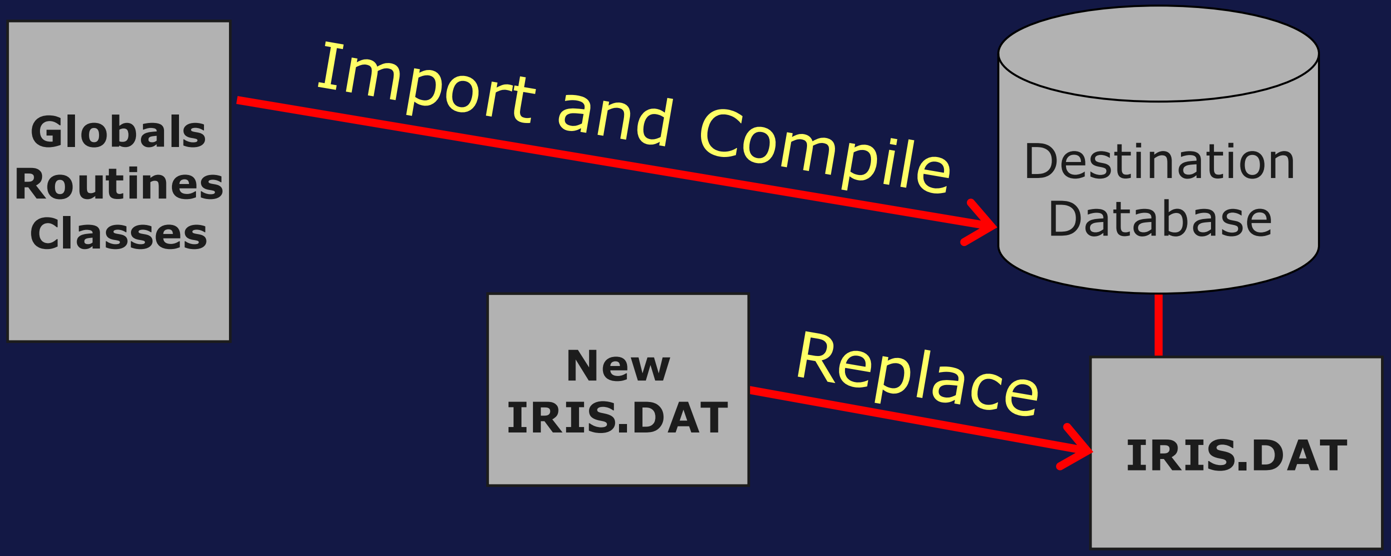
Applications

Overview

- Application installation.
- Installer package.
- Classes overview.
  - Objects.
  - Routines.
- Compiled classes.
- Phonebook application installation.

Application Installation

- Several ways to install application.
  - Receive files containing routines, globals and/or classes.
    - Import and compile in database.
  - Receive IRIS.DAT file containing globals, routines and classes.
    - Dismount database.
    - Replace current IRIS.DAT file with new one.



%Installer Package

- Classes to help automate installation of application.
  - Can control:
    - Namespaces/databases/mappings.
    - System-wide settings.
    - Security settings.
  - Can load code and data.
  - Can execute any custom code.

%Installer Package (cont.)

- Use %Installer manifest to customize install or upgrade:
  - Manifest XML block contains installation details.

```
XData MyInstall [XMLNamespace = INSTALLER]
<Manifest>
  <Namespace>
    <Configuration>
    <Database />
    <Mapping />
    </Configuration>
    <CSPApplication />
    <Import />
  </Namespace>
</Manifest>
```

File Contents

File Extension	Contents	Destination Machine
.XML	Globals, routines and classes.	Import and compile classes and routines, if needed.
.GSA, .GO and .GBL	Globals.	Import.
.RSA, .RO and .RTN	Routines.	Import and compile, if needed.

Importing and Exporting

	Management Portal System Explorer	Command Line
Global	Globals > Export Globals > Import	`%GO, %GI` `%GOF, %GIF` (uses non-ASCII formats and supports control characters.)
Routine	Routines > Export Routines > Import	`%RO, %RI` `%ROMF, %RIMF` (for .OBJ routines)

Class	Classes > Export
	Classes > Import
		`$SYSTEM.OBJ.Export()`
		`$SYSTEM.OBJ.Load()`
		Can also import/export globals and routines using same XML file.

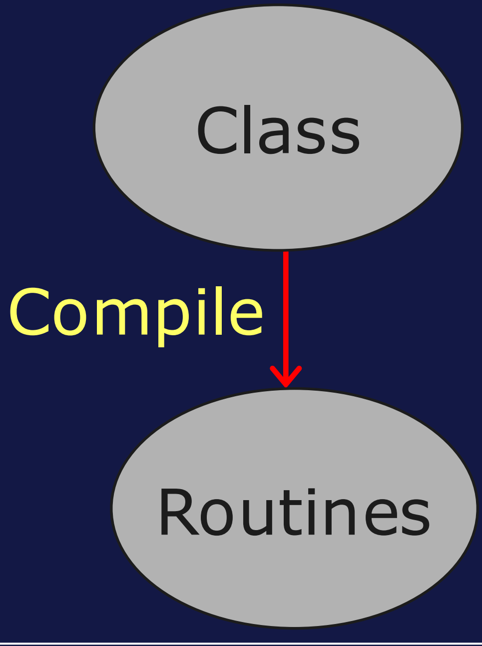
- Can import/export routines and classes using VS Code or Studio.

Classes Overview

- Class.
  - Defines data applications will manipulate.
  - Defines application logic for that data.
- Class definition contains definitions of:
  - Properties.
    - Define data elements of class.
  - Methods.
    - Functions or actions to perform on data.
  - Indexes.
    - Provide fast access to data.
  - And more…

Routine Generation

- Compiling class generates routine definitions to support object and SQL access to data.
  - Routines support object and SQL operations.
    - .INT → .OBJ.
    - Generates table definitions (meta-data) to support SQL access.



Default Routine/Global Names

- Default routines and globals for class:
  - Routines: *PackageName.ClassName.Suffix.*
  - Globals: *PackageName.ClassNameSuffix.*
- Routines for Sample.Person class:
  - Sample.Person.# for .INT and .OBJ.

Globals for Sample.Person Class

Global	What it contains
^Sample.PersonD	Data
^Sample.PersonI	Indexes
^Sample.PersonS	Streams

Suggested Readings

- *Installation Guide > Creating and Using an Installation Manifest.*

Chapter 4: Creating and Using an Installation Manifest

4.1 Overview of an Installation Manifest

4.2 Creating a Class That Defines a Manifest

4.3 Key Options

4.4 Adding Users and Passwords

4.5 Adding Messages

4.6 <Manifest> Tags

4.7 Variables Available within <Manifest>

4.8 Using the Manifest

4.9 Example

Summary

- What are the key points for this module?

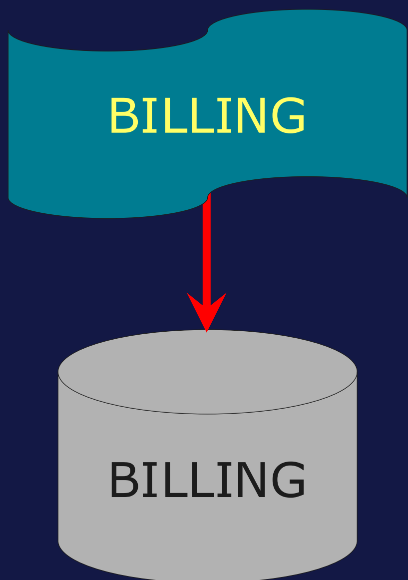
Configuration for the Application

InterSystems
Creative data technology

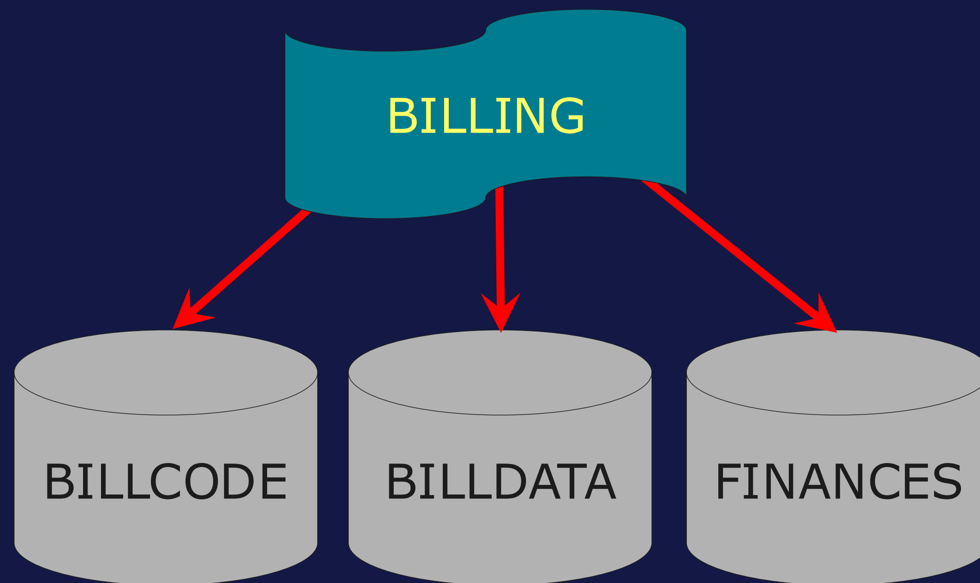
Overview

- Namespace and database configuration.
- Mappings.
- Security configuration.
- Remote System Access.

Namespaces Examples



BILLING: Contains all data and application code.



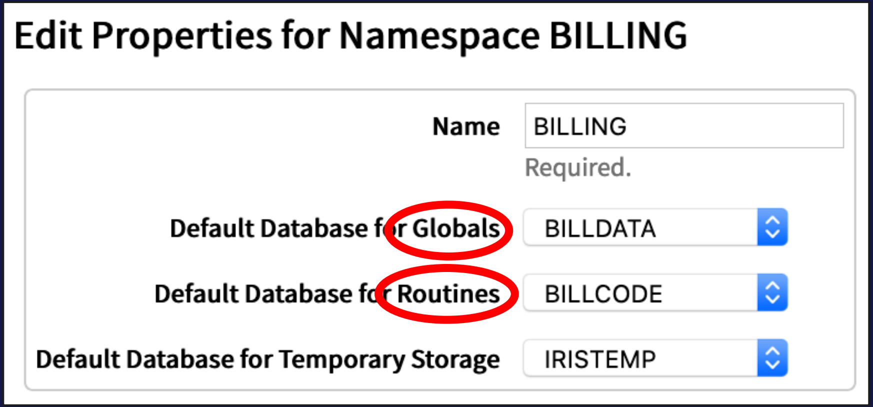
- BILLDATA: Client data.
- BILLCODE: Billing Application code.
- FINANCES: Financial data. (Global Mapping)

Creating Databases and Namespaces

- Using Management Portal.
  - System Administration > Configuration > System Configuration > Namespaces > Create New Namespace.
  - System Administration > Configuration > System Configuration > Local Databases > Create New Database.
    - When creating namespace, can create associated database at same time.
- Using Terminal:
  - `%SYS>do ^NAMESPACE` (v2025.2+)
  - `%SYS>do ^DATABASE`

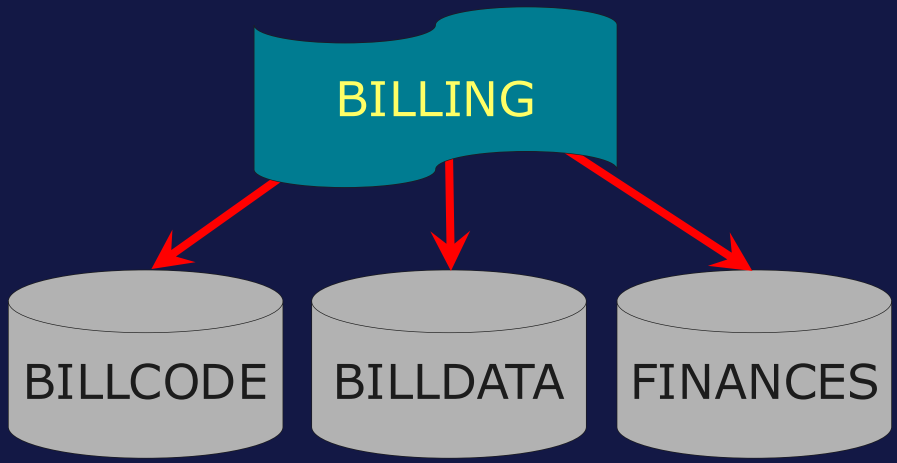
Default Databases for Namespace

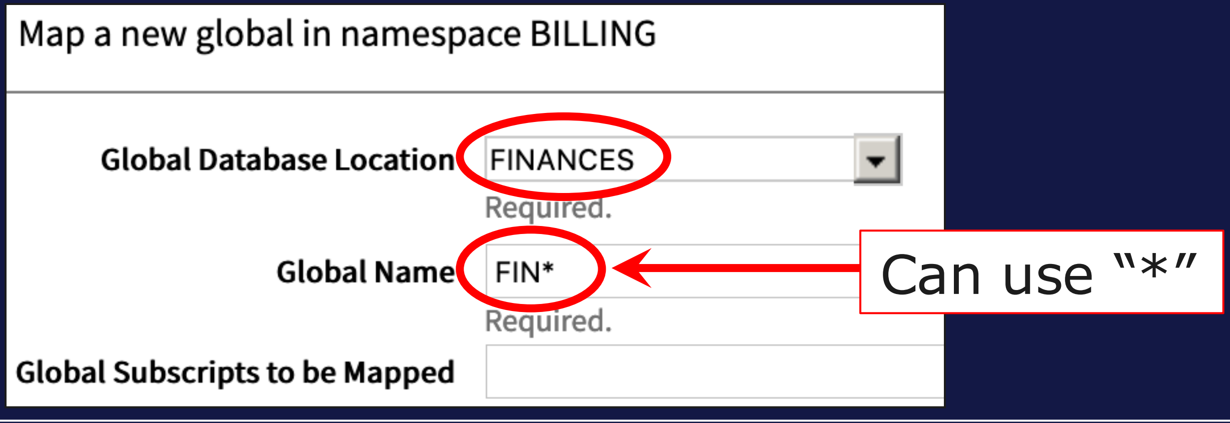
- System Administration > Configuration > System Configuration > Namespaces > Select Namespace.
  - Default Database for Globals:
    - Stores data.
  - Default Database for Routines:
    - Stores code.
    - Also stores code contained in packages of classes.



Mapping Data: Global Mappings

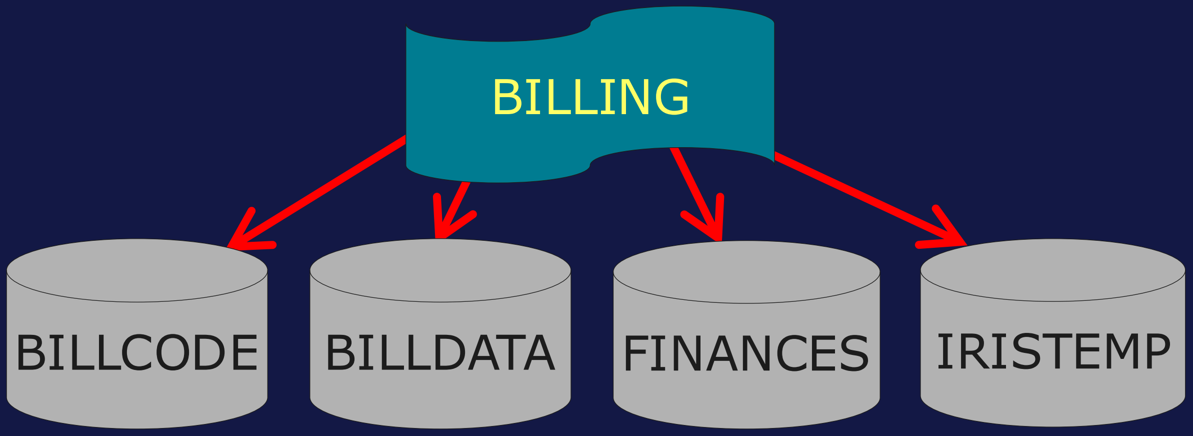
- System Administration > Configuration > System Configuration > Namespaces > Global Mappings.





IRISTEMP Default Global Mappings

- Default global mappings:
  - ^IRIS.Temp*
  - ^CacheTemp*
  - ^mtemp*
- Never change or remove these globals.
  - Used internally by system.
- Can map application’s temporary globals to IRISTEMP.

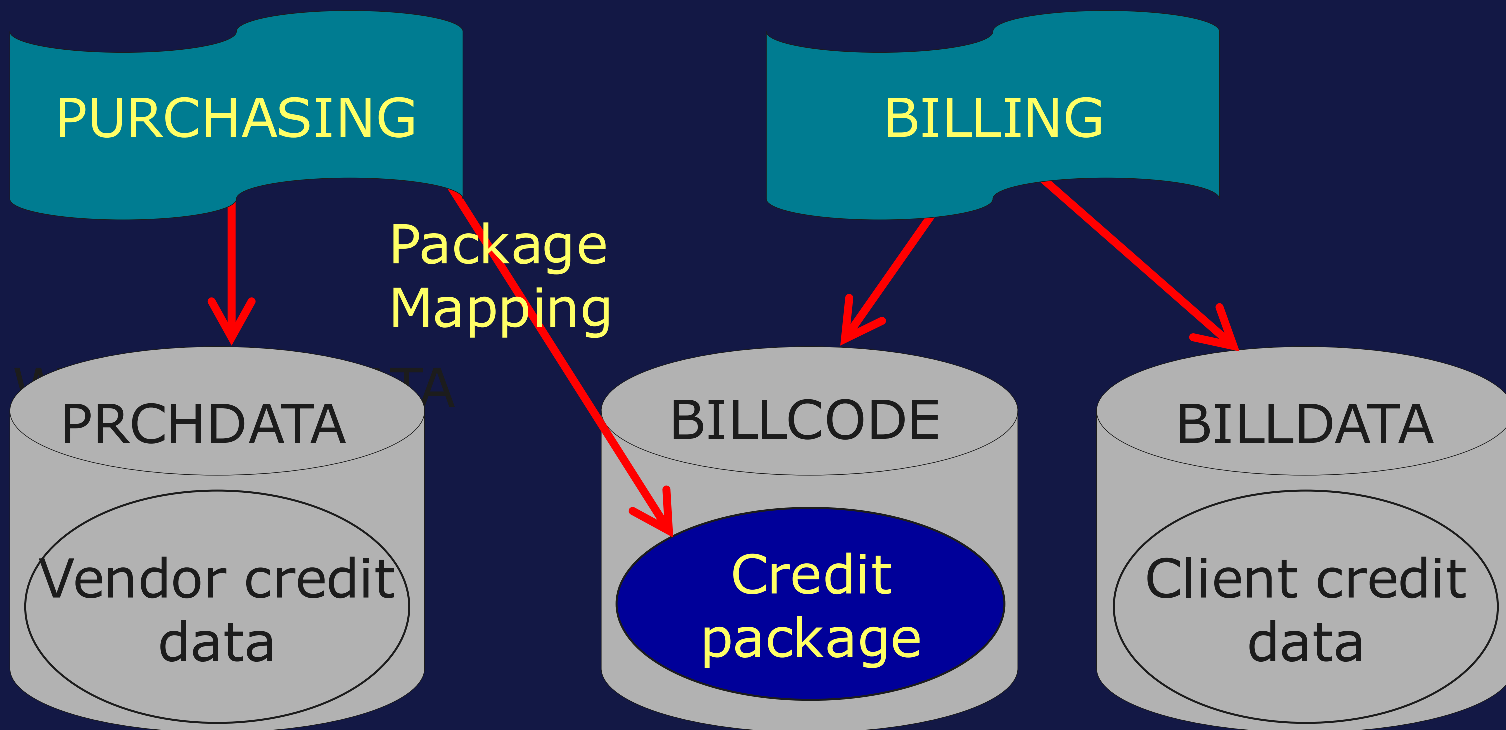


IRISTEMP Features

- Shared by all namespaces.
- Tries to keep data in database cache on server instead of writing to disk.
- Excluded from IRIS online backup.
- Globals never journaled.
- Process-private globals mapped here.
  - ^||*globalname*.
  - Private to current process and automatically deleted at process termination.

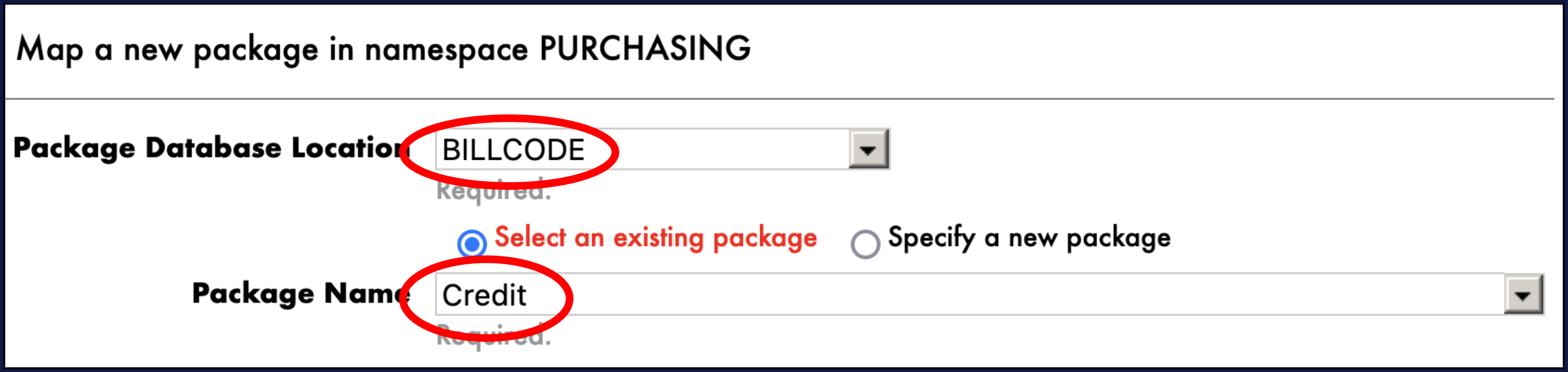
Mapping Code: Package Mappings

- Maps classes and related code.
- Does not map data.
  - Use global mappings if data also needed.



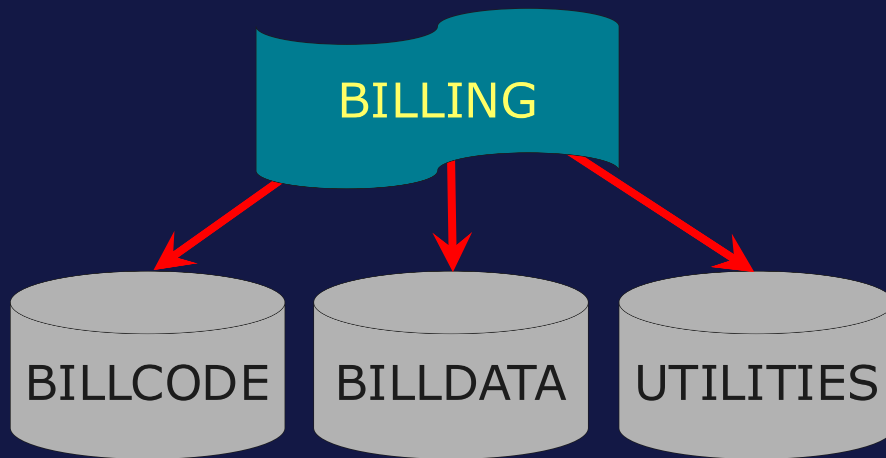
Mapping Code: Package Mappings (cont.)

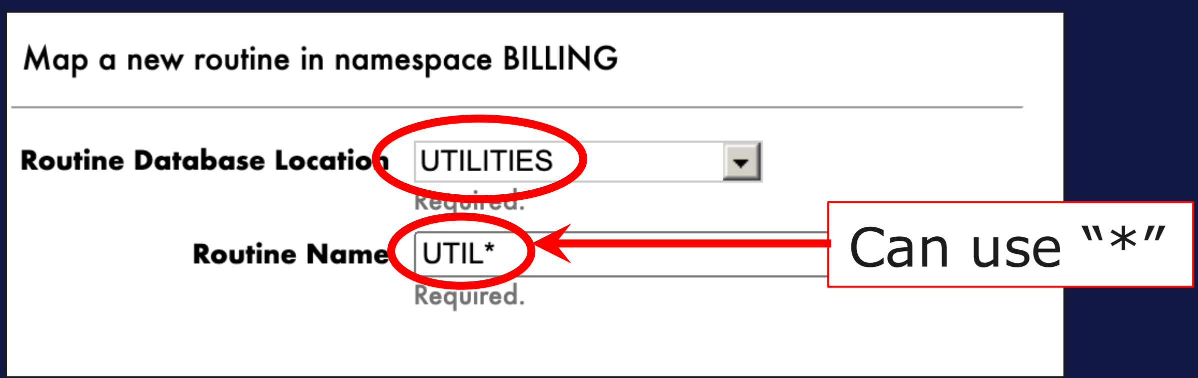
- System Administration > Configuration > System Configuration > Namespaces > Package Mappings.



Mapping Code: Routine Mappings

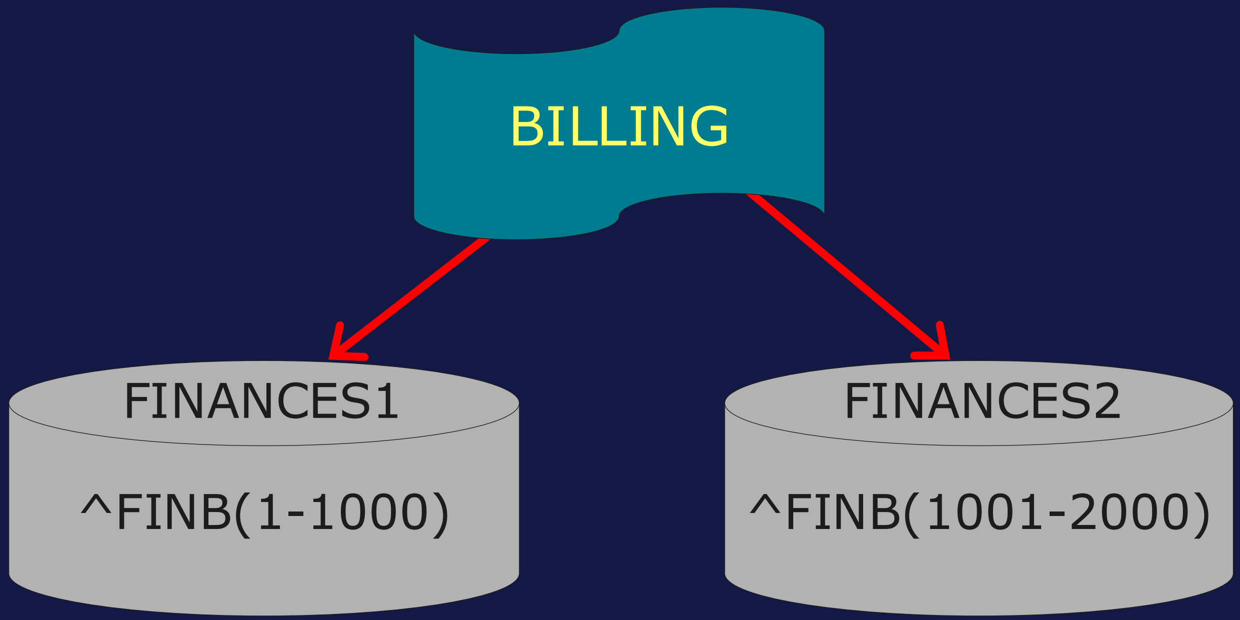
- System Administration > Configuration > System Configuration > Namespaces > Routine Mappings.





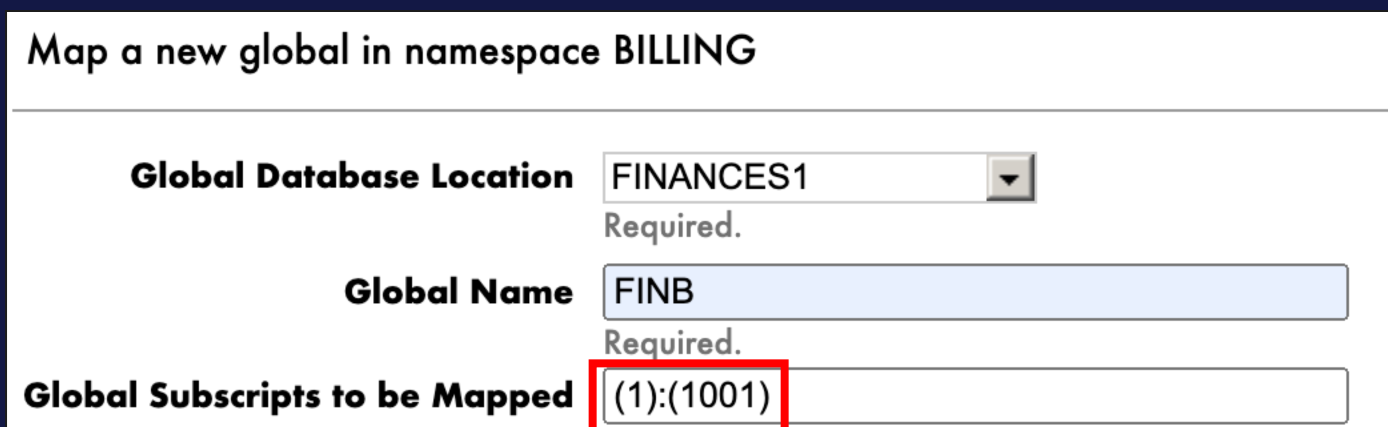
Subscript Level Mapping (SLM)

- Enables namespace to reference **sections** of global from different databases.
  - Used for performance or security.



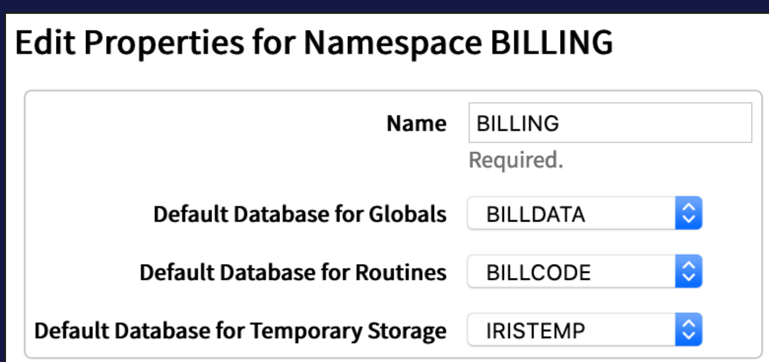
How To: Configure SLM

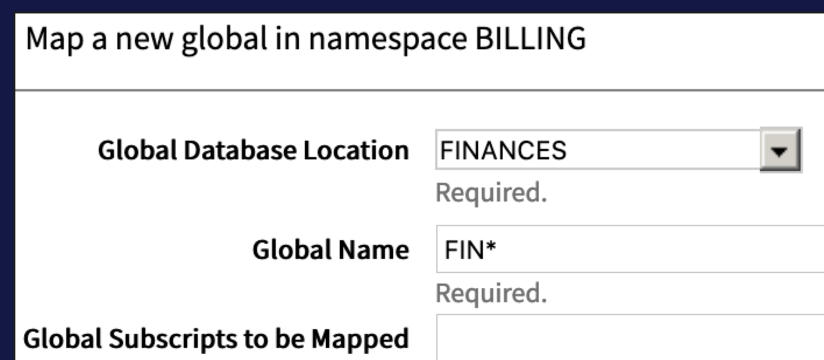
- System Administration > Configuration > System Configuration > Namespaces > Global Mapping > New Global Mappings.
  - Specify database, global and subscripts where mapping will take place.
    - Specifying (1):(1001) means everything starting at (1), up to, but not including (1001).



Mapping Does Not Move Data or Code

- Adding mappings does not move any **existing** data or code from current database to mapped database.
- For example, to move FIN* globals corresponding to the mapping below:
  1. Copy FIN* globals from BILLDATA to FINANCES.
  2. Delete FIN* globals from BILLDATA.





%ALL Namespace

- Create %ALL namespace to map selected data and code to **all** namespaces.
  - Not actual namespace.
- Steps:
  1. Create %ALL namespace.
     - System ignores %ALL’s default databases.
  2. Add desired mappings to %All namespace.
     - Mapped data and code available in **all** namespaces.

Deleting Databases and Namespaces

- Databases.
  - Requires deletion of any namespaces that reference database.
- Namespaces.
  - Does not delete any referenced databases.
- Delete databases and namespaces using Management Portal or Terminal:
  - System Administration > Configuration > System Configuration.
  - `%SYS>do ^DATABASE`
  - `%SYS>do ^NAMESPACE`

Security Configuration

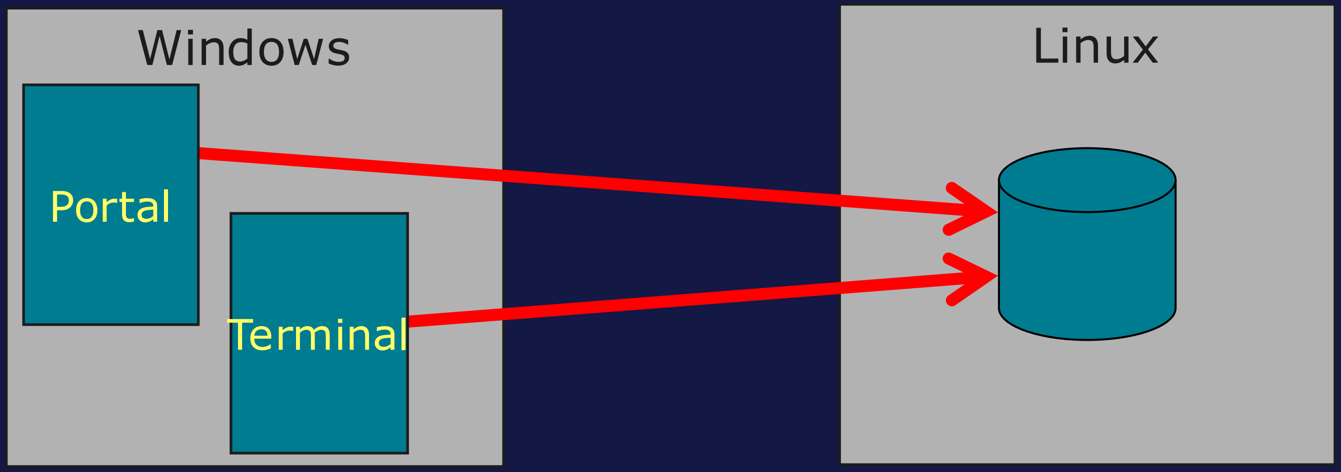
- Create necessary users and roles.
  - System Administration > Security:
    - Users > Create New User.
    - Roles > Create New Role.
- Disable predefined users such as SuperUser and Admin.
  - System Administration > Security:
    - Users > Select user > Clear User enabled.

Security Configuration (cont.)

- Enable or disable services as needed.
  - System Administration > Security:
    - Services > Select service > Select/clear Service enabled.
- Can also use `do ^SECURITY` to edit users, roles, and services.

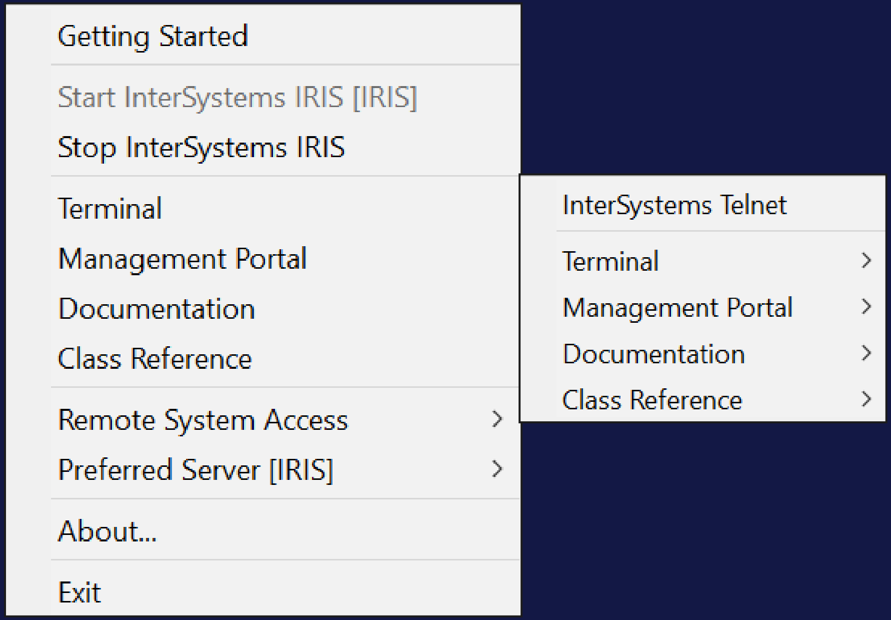
Remote System Access (RSA)

- From current system can access other systems using:
  - Terminal, Management Portal, Studio (pre-v2024.2) and Class Reference.
    - Required for Studio to access non-Windows environment.
  - **Cannot** stop or start remote system using remote system access.



Remote System Access (RSA) (cont.)

- First,
  - Add one or more servers to Preferred Server list.
- Next
  - Use *Remote System Access* option to choose tool.
  - Choose server on which to run it.



Suggested Readings

- *System Administration Guide.*
  - *Configuring InterSystems IRIS > Configuring Data.*
  - *Connecting to Remote Servers.*

Summary

- What are the key points for this module?